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# Incidence, Risk Factors, and Outcomes in Stressor-Associated Atrial Fibrillation: Insights From the VITAL-AF Trial

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**BACKGROUND:** Stressor-associated atrial fibrillation (AF), defined as newly diagnosed AF in the setting of a reversible physiological stressor, is common. However, risk factors, outcomes, and current management practices remain poorly understood.

**METHODS:** We analyzed data from VITAL-AF, a pragmatic, cluster-randomized AF screening trial conducted in 2018 and 2019 comprising adults ≥65 years of age across 16 primary care practices affiliated with Massachusetts General Hospital. All participants had longitudinal follow-up for adjudicated incident AF (including whether stressor-associated versus nonstressor-associated ["primary"]) and clinical outcomes. We compared associations between clinical AF risk factors and incident AF group (stressor-associated versus primary) using Fine-Gray models handling death and each AF group as competing risks. We also quantified oral anticoagulant initiation rates after AF diagnosis. We then fit Cox proportional hazards models to quantify associations between incident AF group (as a time-varying covariate) and a composite end point of major bleeding, stroke, and all-cause mortality, with adjustment for $CHA_2DS_2$-VASc (congestive heart failure, hypertension, age >74, diabetes, stroke or transient ischemic attack or thromboembolism, vascular disease, age 65–74, sex category; stroke) and ATRIA (Anticoagulation and Risk Factors in Atrial Fibrillation; bleeding) scores and time-varying oral anticoagulant exposure.

**RESULTS:** We analyzed 30 265 patients (41% men, 83% White, and mean age 74 years). Of 988 incident AF events, 290 (29%) were stressor associated. Clinical risk factors, including age, hypertension, and heart failure, showed similar associations with both primary and stressor-associated AF. Oral anticoagulant initiation within 90 days of new AF diagnosis was lower for stressor-associated versus primary AF (56% versus 75%, $P<0.001$). The incidence of the composite end point was 2.30 per 100 person-years (95% CI, 2.17–2.44) for no AF, 15.09 (95% CI, 12.22–18.42) for primary AF, and 22.71 (95% CI, 16.91–29.87) for stressor-associated AF. In adjusted models and using no AF as the referent, both primary AF (hazard ratio [HR], 3.91 [95% CI, 2.89–5.28]) and stressor-associated AF (HR, 6.13 [95% CI, 4.31–8.72]) were associated with substantially higher risk of the composite end point.

**CONCLUSIONS:** Among >30 000 primary care patients with adjudicated AF events, nearly one-third of incident AF cases were stressor associated. Despite similar risk factor profiles and comparably high rates of adverse outcomes, oral anticoagulant initiation is lower when AF is stressor associated. Future work is needed to define optimal approaches to prevention, surveillance, and management of stressor-associated AF.

**Key Words:** atrial fibrillation ■ prevention ■ randomized trials ■ risk factors

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## Clinical Perspective

### What Is New?

- In a randomized trial of atrial fibrillation (AF) screening including >30 000 primary care patients, stressor-associated AF accounted for nearly one-third of incident AF diagnoses and was associated with risk factors such as age and heart failure.
- Despite comparably high rates of adverse outcomes to primary AF, patients with stressor-associated AF are less likely to receive oral anticoagulants, even in the context of elevated stroke risk.

### What Are the Clinical Implications?

- Risk factor profiles are similar for both primary and stressor-associated AF, suggesting that guideline-based optimization of modifiable AF risk factors may be effective in preventing stressor-associated AF.
- Opportunities may exist to improve allocation of oral anticoagulants after stressor-associated AF, and future work is warranted to identify optimal candidates.

## Nonstandard Abbreviations and Acronyms

| | |
|---|---|
| **AF** | atrial fibrillation |
| **EHR** | electronic health record |
| **HR** | hazard ratio |
| ***ICD-10*** | *International Classification of Diseases, Tenth Revision* |
| **OAC** | oral anticoagulant |
| **PY** | person-year |

Stressor-associated atrial fibrillation (AF), defined as newly diagnosed AF occurring in the setting of a reversible, acute physiological stressor, is highly prevalent.[1–5] AF stressors span a broad range of exposures, including cardiac and noncardiac surgery, infection, and acute cardiovascular or pulmonary illness.[1,2] Despite longstanding perceptions that stressor-associated AF is benign and self-limited, accumulating evidence suggests that stressor-associated AF is associated with adverse outcomes, including recurrent AF, stroke, and death, at rates potentially approaching those observed for nonstressor-associated or primary AF.[3–5]

However, most existing data on stressor-associated AF derive from smaller retrospective analyses or very specific clinical settings (eg, after cardiac surgery), limiting generalizability.[4,6–12] Larger analyses have commonly relied on billing and diagnostic codes to classify AF type and outcomes, introducing potential misclassification.[4,13] Furthermore, real-world practice patterns surrounding oral anticoagulant (OAC) initiation for stressor-associated AF remain poorly characterized.[14–16] A systematic assessment of risk factor profiles, OAC use, and AF-related outcomes in a large prospective sample of individuals with clinically adjudicated AF events spanning a broad range of stressors is needed to inform future efforts to optimally prevent and manage stressor-associated AF.

Here, we leveraged data from the VITAL-AF trial, a large pragmatic cluster-randomized AF screening trial enrolling >30 000 older adults receiving primary care with clinically adjudicated AF, stroke, and bleeding events, to characterize risk factors, management strategies, and outcomes after stressor-associated AF.[17,18]

# METHODS

## Data Sharing

VITAL-AF data contain protected health information and cannot be shared publicly. Data will be made available under a data use agreement upon reasonable request by the corresponding author.

## Study Design

The design, conduct, and primary outcome results for the VITAL-AF trial have been published previously.[17,18] Briefly, VITAL-AF was a pragmatic cluster-randomized trial of AF screening using a handheld single-lead ECG device embedded in primary care practices. Sixteen Massachusetts General Hospital practices were randomized to either AF screening or usual care. Eligible patients were ≥65 years of age and had an outpatient clinic appointment at a participating primary care practice with a primary care physician, nurse practitioner, or physician's assistant. Enrollment took place between July 31, 2018, and October 8, 2019. Participants were followed for clinical outcomes for up to 2 years. Given the pragmatic nature of VITAL-AF, individuals with prevalent AF were enrolled; however, in the current analysis, we focused only on individuals without prevalent AF. This study adhered to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines (Supplemental Material). The VITAL-AF study was approved by the Mass General Brigham institutional review board. Participants provided informed consent to participate.

## Ascertainment of Stressor-Associated AF

AF ascertainment followed a tiered protocol. Candidate AF events were first identified using a high-sensitivity electronic health record (EHR)–based algorithm comprising *International Classification of Diseases, Tenth Revision* (*ICD-10*), codes and 12-lead ECGs.[17,19] All potential incident AF events identified by the algorithm then underwent manual adjudication by 2 independent research nurses who reviewed all relevant components of the EHR. Any discrepancies in AF ascertainment were resolved through consensus or, in cases of continued uncertainty, by a study cardiologist.

Stressors were identified based on previous literature and categorized into clinically related groups as follows: cardiac surgery, noncardiac surgery (thoracic and nonthoracic surgery), infection (sepsis, pneumonia, and other infections), respiratory illness (acute pulmonary disease and pulmonary embolism),

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cardiac disease (myocardial infarction, pericarditis, myocarditis, and cardiac arrest), and toxicoses (thyrotoxicosis and acute alcohol intoxication).[1,2,4] In accordance with the VITAL-AF protocol, study nurses conducted an independent double review of the medical record for stressor events within 30 days preceding each incident AF diagnosis.[1,2,17] Discrepancies in stressor adjudication were resolved post hoc by a study cardiologist (J.H.). All incident AF events not meeting criteria for a stressor-associated AF episode were considered nonstressor-associated or primary AF. Further details of the stressor-associated AF adjudication process, including the relevant case report form, are provided in the Supplemental Material. In keeping with previous evidence demonstrating the potential for onset of stressor-associated AF contemporaneously with a stressor, we did not institute any required lag time between stressor and AF onset.[5,20–22]

## Exposures and Outcomes

Baseline demographics and clinical factors were ascertained at the time of enrollment using EHR data. Clinical factors of interest included all components of the validated CHARGE (Cohorts for Heart and Aging Research in Genomic Epidemiology) AF risk score (age, race, systolic and diastolic blood pressure, height, weight, anti-hypertensive medication use, myocardial infarction, current smoking, and heart failure),[23] additional components of the $CHA_2DS_2$-VASc (congestive heart failure, hypertension, age >74, diabetes, stroke or transient ischemic attack or thromboembolism, vascular disease, age 65–74, sex category) stroke score (sex, previous stroke, or vascular disease), and the ATRIA (Anticoagulation and Risk Factors in Atrial Fibrillation) bleeding score (anemia, renal disease, or previous bleeding).[24] OAC exposure was ascertained using EHR-based medication lists, which reflect prescriptions and provider-documented medication entries and included apixaban, dabigatran, edoxaban, rivaroxaban, and warfarin.

The primary outcome was a composite of prespecified VITAL-AF trial outcomes comprising major bleeding, stroke, and all-cause mortality. Secondary outcomes included a composite of stroke or all-cause mortality as well as each individual component outcome. Individual component outcomes were subject to limited event rates and should be considered exploratory. All stroke and bleeding events were adjudicated by the VITAL-AF clinical end point committee.[17] Mortality was ascertained using EHR data and the Social Security Death Index.

## Statistical Analysis

To quantify risk factor profiles for stressor-associated versus primary AF, we fit multivariable Fine-Gray subdistribution hazard models with incident AF group (ie, stressor-associated versus primary) and death modeled as competing risks. Person-time began at the first study visit and continued until the earliest time of AF diagnosis, end of study follow-up, or death. Models were adjusted for each individual component of CHARGE-AF as well as sex.[23] Given minimal missingness (n=450, 1.5%), a complete case analysis was performed (Figure S1). To assess whether the association of a given covariate with AF differed on the basis of AF group (primary AF and stressor-associated AF), we compared the coefficient of that covariate for primary AF with its coefficient for stressor-associated AF, estimated the standard error of that difference using 1000-iteration bootstrapping, and used this estimate to generate a 95% CI and conduct a 2-sided *Z* test. For interpretability, we report the difference between coefficients on the ratio scale, corresponding to a percentage change in relative hazard (eg, a hazard ratio [HR] of 1.50 for stressor-associated AF versus an HR of 1.25 for primary AF would correspond to a ratio of relative hazards of 1.50/1.25=1.33 or 33% higher HR for stressor-associated AF). In secondary models, we additionally adjusted for exposure to AF screening.

We then sought to quantify and compare rates of OAC prescriptions as a function of incident AF group. Prevalent OAC exposure was defined as any OAC use before the AF diagnosis date. New OAC initiation was defined as a new exposure within 90 days of AF diagnosis. To evaluate whether OAC use varied by stroke risk, we examined new OAC initiation rates by $CHA_2DS_2$-VASc score categories aligned with guideline-based thresholds for OAC use in AF: medium risk (3 to 4 in women and 2 to 3 in men) and high risk (≥5 in women and ≥4 in men).[2] We compared rates of OAC use between AF groups, stratified by risk category, using the $\chi^2$ test. We also assessed for differences in baseline characteristics by OAC use among individuals with stressor-associated AF using the $\chi^2$ test.

We then tabulated the number of individuals with evidence of subsequent AF detection after the index incident AF diagnosis, defined as diagnostic or procedural evidence of AF. AF was identified from 12-lead ECG and patch or event monitor diagnostic statements or the presence of billing codes corresponding to synchronized direct-current cardioversion (Current Procedure Terminology codes 92960 and 92961, and *ICD-9* code 99.61). To ensure that subsequent AF detection was discrete from the index AF episode, diagnostic tests and direct-current cardioversion occurring within 14 days of initial AF diagnosis were excluded. AF pattern was classified as paroxysmal or persistent based on the presence of patch or event monitor reports. AF detection rates and AF pattern were compared between AF groups using $\chi^2$ and Fisher exact tests.

We next sought to compare the incidence of adverse clinical outcomes after primary versus stressor-associated AF. In this analysis, person-time began at incident AF diagnosis (incident AF) or first study visit (no AF) and spanned until the earliest of the respective outcome event, end of study follow-up, or death. Outcomes occurring on the same day as the index AF episode were considered incident. A small number of individuals (36 with primary AF and 11 with stressor-associated AF) with an outcome event preceding incident AF diagnosis were included in the no AF group. Cumulative incidences were estimated using the cumulative incidence function with death as a competing risk for major bleeding and stroke, whereas the remaining outcomes (ie, those including death as a component) were estimated using the Kaplan-Meier method.

We then quantified associations of stressor-associated and primary AF with adverse outcomes using Cox proportional hazards models, treating both incident AF group and OAC exposure as time-varying covariates. Follow-up time began at the first study visit, and prevalent OAC use was defined as exposure before the first study visit. Models were adjusted for the continuous $CHA_2DS_2$-VASc and ATRIA scores. In a sensitivity analysis, Cox models were refit, adjusting for the unique components of the $CHA_2DS_2$-VASc and ATRIA scores (congestive heart failure, age, hypertension, diabetes, female sex, previous stroke, vascular disease, anemia, renal dysfunction, and previous hemorrhage) and time-varying OAC use. In an additional sensitivity analysis, we assessed a more stringent exposure definition

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in which only documented OAC exposure within the year before the first study visit was considered prevalent OAC use.

We additionally performed an exploratory analysis to assess whether the association of OAC exposure with outcomes varied by AF group. Here, we fit Cox proportional hazards restricted to individuals with incident AF and additionally including an interaction term between AF group (primary versus stressor associated) and time-varying OAC exposure. Follow-up time began at AF diagnosis and spanned until the earliest of the respective outcome event, end of study follow-up, or death.

We considered 2-sided $P<0.05$ to indicate statistical significance. For all time-to-event models, we observed no substantive deviations from proportional hazards, as assessed through visual examination of model residuals versus time, as well as tests of interaction between covariates and time. The functional form of continuous covariates was assessed by fitting penalized splines, which supported the validity of a linear fit in all cases. Analyses were performed in Python 3.12.4 and R 4.4.0.

# RESULTS

## Study Sample

The analysis sample comprised 30 265 individuals without prevalent AF, of whom 41% were men, 83% were White, and the mean age was 74 (SD 6.8) years (Table 1; Figure S1). Over a median of 1.7 years of follow-up (quartile 1, 1.5; quartile 3, 1.9), incident AF occurred in 988 (3.3%) individuals, of which 698 (71%) were classified as primary AF and 290 (29%) were classified as stressor-associated AF. Among those with stressor-associated AF, infection (n=115, 40%) was the most common stressor, followed by cardiac surgery (n=76, 26%), noncardiac surgery (n=47, 16%), cardiac disease (n=30, 10%), respiratory illness (n=16, 6%) and toxicoses (n=6, 2%).

## Associations of Clinical Factors With Incident AF

The baseline CHARGE-AF score was similar between those who developed primary AF (mean 14.3, SD 1.0) and stressor-associated AF (mean 14.2, SD 1.0) but lower among those who did not develop AF (mean 13.4, SD 0.9, $P<0.001$ compared with either AF group; Table 1). In general, risk factor profiles were similar for stressor-associated and primary AF (Figure 1). Several established AF risk factors showed strong associations

**Table 1. Baseline Characteristics**

| Characteristic | Overall (n=30 265) | No AF (n=29 277) | Primary AF (n=698) | Stressor-associated AF (n=290) |
|---|---|---|---|---|
| Age, y | 73.9 (6.8) | 73.8 (6.7) | 78.2 (8.0) | 77.2 (7.6) |
| Male sex | 12 431 (41.1%) | 11 907 (40.7%) | 356 (51%) | 168 (57.9%) |
| White race | 25 405 (82.7%) | 24 520 (82.5%) | 627 (88.3%) | 258 (88.1%) |
| Height, cm | 164.8 (10.5) | 164.8 (10.4) | 166.4 (11.6) | 167.0 (10.7) |
| Weight, kg | 76.1 (17.7) | 75.9 (17.7) | 80.8 (20.0) | 79.0 (18.4) |
| Systolic blood pressure, mm Hg | 131.2 (16.6) | 131.1 (16.5) | 133.0 (17.4) | 133.8 (18.4) |
| Diastolic blood pressure, mm Hg | 74.4 (8.6) | 74.5 (8.6) | 72.8 (9.5) | 72.8 (8.8) |
| Heart rate, bpm | 74.6 (12.9) | 74.6 (12.8) | 73.6 (15.1) | 73.7 (13.4) |
| Current smoker | 1507 (4.9%) | 1454 (4.9%) | 33 (4.6%) | 20 (6.8%) |
| Hypertension | 23 092 (75.2%) | 22198 (74.7%) | 629 (88.6%) | 265 (90.4%) |
| Myocardial infarction | 2170 (7.1%) | 2040 (6.9%) | 93 (13.1%) | 37 (12.6%) |
| Coronary artery disease | 6356 (20.7%) | 6001 (20.2%) | 247 (34.8%) | 108 (36.9%) |
| Heart failure | 3158 (10.3%) | 2923 (9.8%) | 166 (23.4%) | 69 (23.5%) |
| Vascular disease | 5962 (19.4%) | 5614 (18.9%) | 246 (34.6%) | 102 (34.8%) |
| Previous stroke | 2560 (8.3%) | 2420 (8.1%) | 106 (14.9%) | 34 (11.6%) |
| Renal disease | 4508 (14.7%) | 4252 (14.3%) | 171 (24.1%) | 85 (29%) |
| Anemia | 7534 (24.5%) | 7167 (24.1%) | 259 (36.5%) | 108 (36.9%) |
| Previous bleeding | 11 485 (37.4%) | 11 085 (37.3%) | 286 (40.3%) | 114 (38.9%) |
| Hypertension medication | 21 437 (69.8%) | 20 587 (69.3%) | 606 (85.4%) | 244 (83.3%) |
| Oral anticoagulants | 703 (2.3%) | 667 (2.2%) | 27 (3.8%) | 9 (3.1%) |
| CHARGE-AF score | 13.5 (1.0) | 13.4 (0.9) | 14.3 (1.0) | 14.2 (1.0) |
| $CHA_2DS_2$-VASc score | 3.4 (1.4) | 3.4 (1.4) | 4.2 (1.6) | 4.1 (1.5) |
| ATRIA bleeding score | 3.1 (2.5) | 3.0 (2.5) | 4.3 (2.7) | 4.4 (2.6) |

Continuous data are presented as mean (SD), and categorical data are presented as n (%). AF indicates atrial fibrillation; ATRIA, Anticoagulation and Risk Factors in Atrial Fibrillation; $CHA_2DS_2$-VASc, congestive heart failure, hypertension, age >74, diabetes, stroke or transient ischemic attack or thromboembolism, vascular disease, age 65-74, sex category; and CHARGE, Cohorts for Heart and Aging Research in Genomic Epidemiology.

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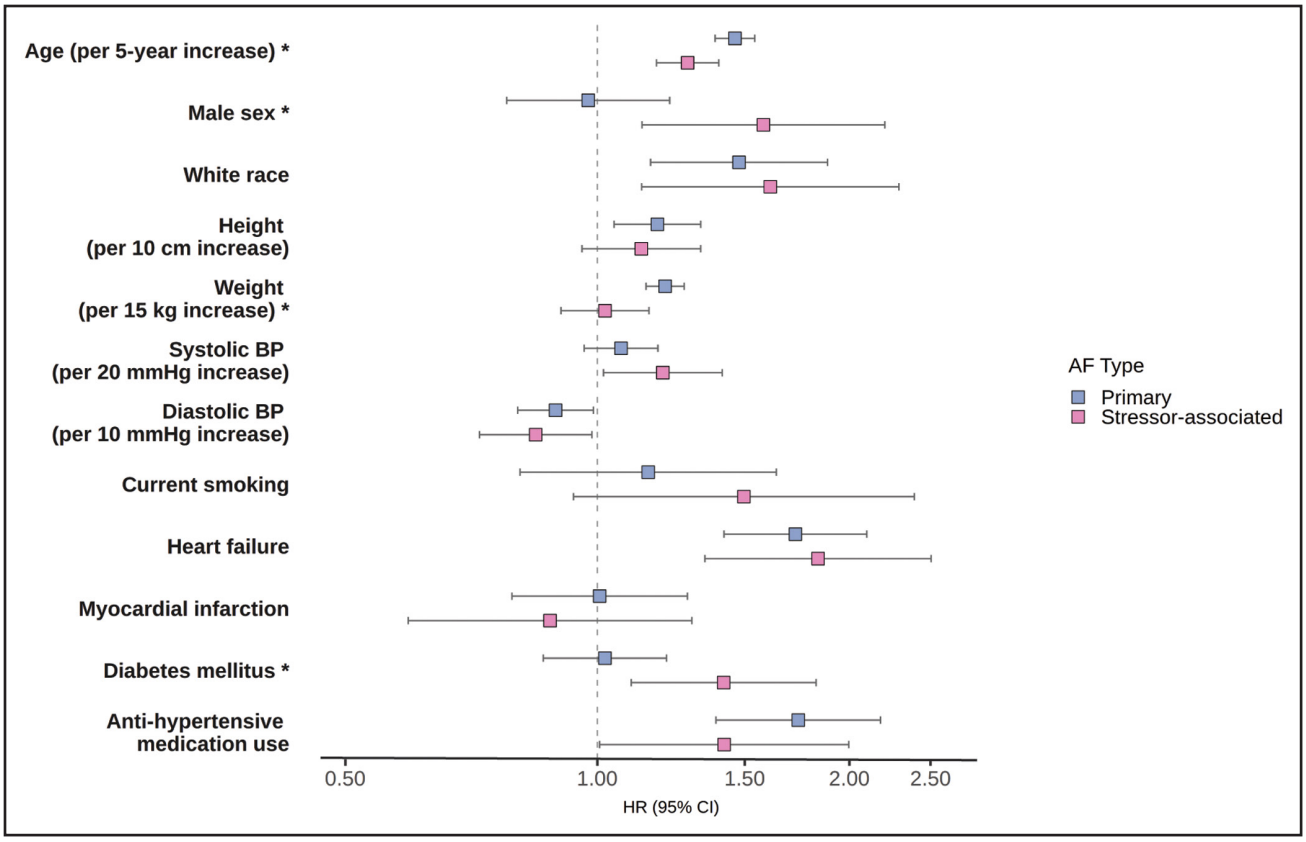


**Figure 1. Associations of atrial fibrillation (AF) risk factors with incident AF.**
Associations of sex, clinical factors included in the CHARGE (Cohorts for Heart and Aging Research in Genomic Epidemiology) AF clinical risk score (age, White race, height, weight, systolic and diastolic blood pressure, current smoking, heart failure, myocardial infarction, diabetes, and antihypertensive medication use) and incident primary and stressor-associated AF. Subdistribution hazard ratios and 95% CIs are shown for primary and stressor-associated AF (purple and blue, respectively). Asterisks indicate a significant difference in the ratio of relative hazards for primary vs stressor-associated AF (see main text). BP indicates blood pressure; and HR, hazard ratio.

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with higher risk of both primary and stressor-associated AF, including age (primary HR, 1.47 per 5-year increase [95% CI, 1.39–1.55]; stressor-associated HR, 1.29 [95% CI, 1.19–1.41]), heart failure (primary HR, 1.75 [95% CI, 1.44–2.13]; stressor-associated HR, 1.87 [95% CI, 1.37–2.55]), and White race (primary HR, 1.47 [95% CI, 1.15–1.87]; stressor-associated HR, 1.60 [95% CI, 1.12–2.28]; Figure 1; Table S1). When comparing effect sizes for specific risk factors directly, older age (14% higher HR per 5-year increase, $P$=0.01) and greater weight (19% higher HR per 15-kg increase, $P$=0.02) were more strongly associated with primary AF, whereas male sex (61% higher HR, $P$=0.02) and diabetes (39% higher HR, $P$=0.03) were more strongly associated with stressor-associated AF (Table S1). Risk factor profiles were similar with additional adjustment for exposure to AF screening (Table S2).

## OAC Use After Incident AF Diagnosis

Prevalent OAC use was present in 1053 (4%) individuals who did not develop AF, 51 (7%) who developed primary AF, and 24 (8%) who developed stressor-associated AF. Among individuals without prevalent OAC use, new initiation within 90 days of incident AF diagnosis occurred in 484 (75%) after primary AF and 147 (56%) after stressor-associated AF ($P$<0.001 versus primary AF). Compared with primary AF, rates of new OAC initiation within 90 days remained lower among individuals with stressor-associated AF at elevated stroke risk, according to $CHA_2DS_2$-VASc, in both the medium-risk (58% versus 78%, $P$<0.001) and high-risk categories (52% versus 71%, $P$<0.001; Figure 2). Rates of new OAC initiation by stressor are shown in Figure S2. Among patients with stressor-associated AF with no prevalent OAC exposure, those with older age ($P$=0.03), non-White race ($P$=0.048), renal insufficiency ($P$=0.01), or previous anemia ($P$=0.01) were less likely to have been initiated on OACs within 90 days of AF diagnosis (Table S3).

## Subsequent AF Detection

Frequency of ECG, patch monitor, or direct-current cardioversion after index AF diagnosis was identical between individuals with primary AF (63%) and stressor-associated AF (63%; Table S4). Among individuals with

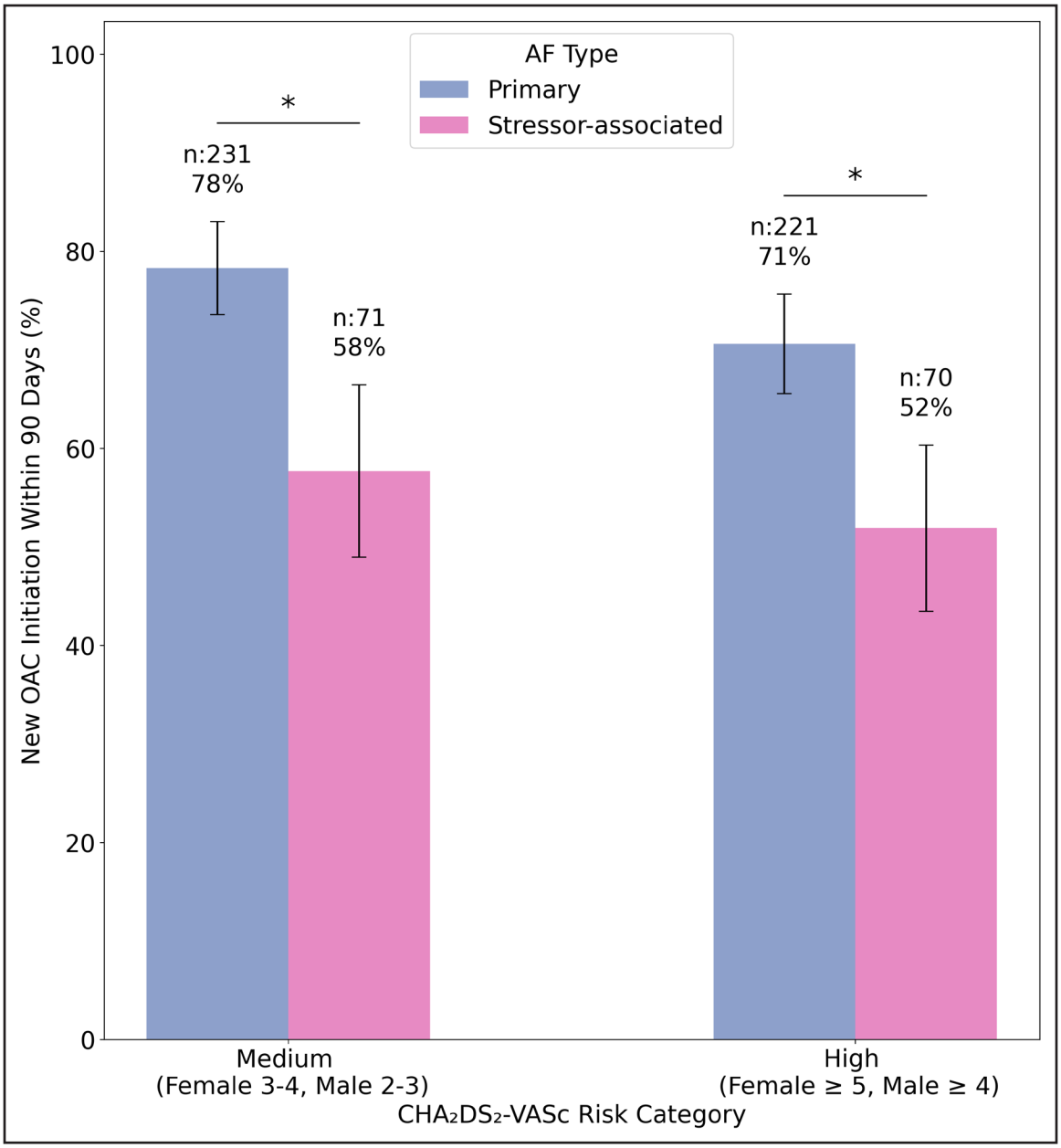


**Figure 2. New oral anticoagulation initiation within 90 days of primary versus stressor-associated atrial fibrillation (AF) diagnosis by $CHA_2DS_2$-VASc risk.**

Shown is the percentage of individuals with new oral anticoagulant initiation within 90 days of AF diagnosis, stratified by medium (women, 3 to 4; and men, 2 to 3), and high (women, ≥5 and men, ≥4) $CHA_2DS_2$-VASc risk category and incident AF group (primary AF and stressor-associated AF). Bars represent the proportion of individuals with new oral anticoagulant exposure within 90 days, with 95% CIs. The exact percentage and corresponding sample size (n) are displayed above each bar. Asterisks denote statistically significant differences between groups ($P<0.05$, $\chi^2$ test). Oral anticoagulant initiation was consistently higher among individuals with primary AF compared with those with stressor-associated AF across both risk strata. $CHA_2DS_2$-VASc indicates congestive heart failure, hypertension, age >74, diabetes, stroke or transient ischemic attack or thromboembolism, vascular disease, age 65-74, sex category; and OAC, oral anticoagulant.

≥1 ECG, patch, or event monitor or direct-current cardioversion, subsequent AF detection was more common after primary AF (53%) versus stressor-associated AF (30%, $P<0.05$). Among patients who underwent patch or event monitoring allowing classification of AF pattern, persistent AF was more frequently observed after primary AF (27%) versus stressor-associated AF (13%, $P<0.05$).

## Adverse Outcomes After Incident AF

The primary composite end point of major bleeding, stroke, or all-cause mortality occurred in 1157 individuals with no AF (2.39 events per 100 person-years [PY; 95% CI, 2.26–2.53]), 96 individuals with primary AF (15.09 events per 100 PY [95% CI, 12.22–18.42]), and 51 individuals with stressor-associated AF (22.71 events

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per 100 PY [95% CI, 16.91–29.87]). The corresponding 2-year cumulative incidence rates of the primary composite end point were 4.8% (95% CI, 4.5–5.2) in the no AF group, 20.0% (95% CI, 16.4–24.5) in the primary AF group, and 21.2% (95% CI, 16.4–27.5) in the stressor-associated AF group (Table 2; Figure 3). The secondary composite end point of stroke or all-cause mortality occurred in 946 individuals with no AF (1.95 events per 100 PY [95% CI, 1.83–2.08]), 84 individuals with primary AF (12.97 events per 100 PY [95% CI, 10.34–16.05]), and 49 individuals with stressor-associated AF (21.64 events per 100 PY [95% CI, 16.01–28.61]), corresponding to 2-year cumulative incidence rates of 3.8% (95% CI, 3.5–4.2), 17.7% (95% CI, 14.2–22.0), and 21.7% (95% CI, 15.9–29.8), respectively (Figure 3). Cumulative incidence curves for the individual secondary outcomes are shown in Figures S3 through S5.

## Associations of AF Group and OAC Exposure With Adverse Outcomes After Incident AF

Both $CHA_2DS_2$-VASc stroke and ATRIA bleeding scores were similarly higher in individuals who developed primary AF (mean of 4.2 [SD 1.6] and 4.3 [SD 2.7], respectively) and stressor-associated AF (mean of 4.1 [SD 1.5] and 4.4 [SD 2.6], respectively) compared with those who did not develop AF (mean of 3.4 [SD 1.4] and 3.0 [SD 2.5], respectively; $P<0.001$ for all pairwise comparisons with no AF; Table 1). In models adjusting for $CHA_2DS_2$-VASc and ATRIA scores as well as time-varying exposure to OACs, both primary AF (HR, 3.91 [95% CI, 2.89–5.28]) and stressor-associated AF (HR, 6.13 [95% CI, 4.31–8.72]) were associated with substantially higher risk of the composite end point of major bleeding, stroke, and all-cause mortality versus no AF (Table 2). Similarly, both incident AF groups were associated with higher risk of the secondary composite end point of stroke or all-cause mortality (primary AF HR, 4.19 [95% CI, 3.02–5.81]); stressor-associated AF HR, 7.19 [95% CI, 4.96–10.43]) versus no AF. Sensitivity analyses including individual $CHA_2DS_2$-VASc and ATRIA components instead of the continuous scores, and restricting prevalent OAC exposure to the preceding year yielded results consistent with the primary analyses (Tables S5 and S6).

## Associations of OAC Exposure With Adverse Outcomes by AF Group

We performed an exploratory analysis comparing the association of OAC exposure by incident AF group, observing that OAC exposure was associated with a lower risk of the composite end point (HR, 0.44 [95% CI, 0.29–0.67]) and stroke or all-cause mortality (HR, 0.39 [95% CI, 0.25–0.61]) after primary AF. Effect sizes were similar for stressor-associated AF, but estimates had limited precision (HR, 0.47 [95% CI, 0.15–1.45] and HR, 0.48 [95% CI, 0.15–1.56], respectively; Table S7). Interaction testing did not indicate the presence of heterogeneity in the effect of OACs as a function of AF type for any of the outcomes.

# DISCUSSION

In this analysis of the VITAL-AF trial comprising >30 000 primary care patients ≥65 years of age with manually adjudicated incident AF and clinical outcome

**Table 2. Outcome Events According to AF Group**

| | No AF | | | Primary AF | | | | Stressor-associated AF | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Outcome | Events (n) | Incidence rate* (95% CI) | 2-y cumulative incidence (%)† | Events (n) | Incidence rate* (95% CI) | 2-y cumulative incidence (%)† | HR vs no AF (95% CI) | Events (n) | Incidence rate* (95% CI) | 2-y cumulative incidence (%)† | HR vs no AF (95% CI) |
| Composite end point (stroke, major bleeding, all-cause mortality) | 1157 | 2.39 (2.26–2.53) | 4.8 (4.5–5.2) | 96 | 15.09 (12.22–18.42) | 20.0 (16.4–24.5) | 3.91 (2.89–5.28)‡ | 51 | 22.71 (16.91–29.87) | 21.2 (16.4–27.5) | 6.13 (4.31–8.72)‡ |
| Stroke or all-cause mortality | 946 | 1.95 (1.83–2.08) | 4.0 (3.6–4.3) | 84 | 12.97 (10.34–16.05) | 17.7 (14.2–22.0) | 4.19 (3.02–5.81)‡ | 49 | 21.64 (16.01–28.61) | 21.7 (15.9–29.8) | 7.19 (4.96–10.43)‡ |
| Major bleeding | 278 | 0.57 (0.51–0.64) | 0.9 (0.8–1.1) | 19 | 2.93 (1.76–4.57) | 2.7 (1.5–4.7) | 2.42 (1.31–4.47)‡ | 6 | 2.65 (0.97–5.76) | 1.40 (0.3–6.6) | 2.52 (1.05–6.04)‡ |
| Stroke | 208 | 0.43 (0.37–0.49) | 0.8 (0.6–1.1) | 18 | 2.78 (1.65–4.39) | 3.6 (2.0–6.3) | 8.64 (4.36–17.12)‡ | 6 | 2.65 (0.97–5.77) | 1.10 (0.3–3.3) | 7.62 (3.09–18.81)‡ |
| All-cause mortality | 767 | 1.57 (1.46–1.69) | 3.2 (2.9–3.4) | 68 | 10.27 (7.97–13.02) | 14.2 (11.2–18.1) | 3.49 (2.43–5.00)‡ | 46 | 20.08 (14.70–26.78) | 21.1 (15.2–29.3) | 7.21 (4.8810.67)‡ |

AF indicates atrial fibrillation; and HR, hazard ratio.

*Per 100 person-years.

†Cumulative incidence function (with death as a competing risk) was used for major bleeding and stroke analyses; Kaplan-Meier estimates were used for all other outcomes.

‡Indicates statistical significance ($P<0.05$).

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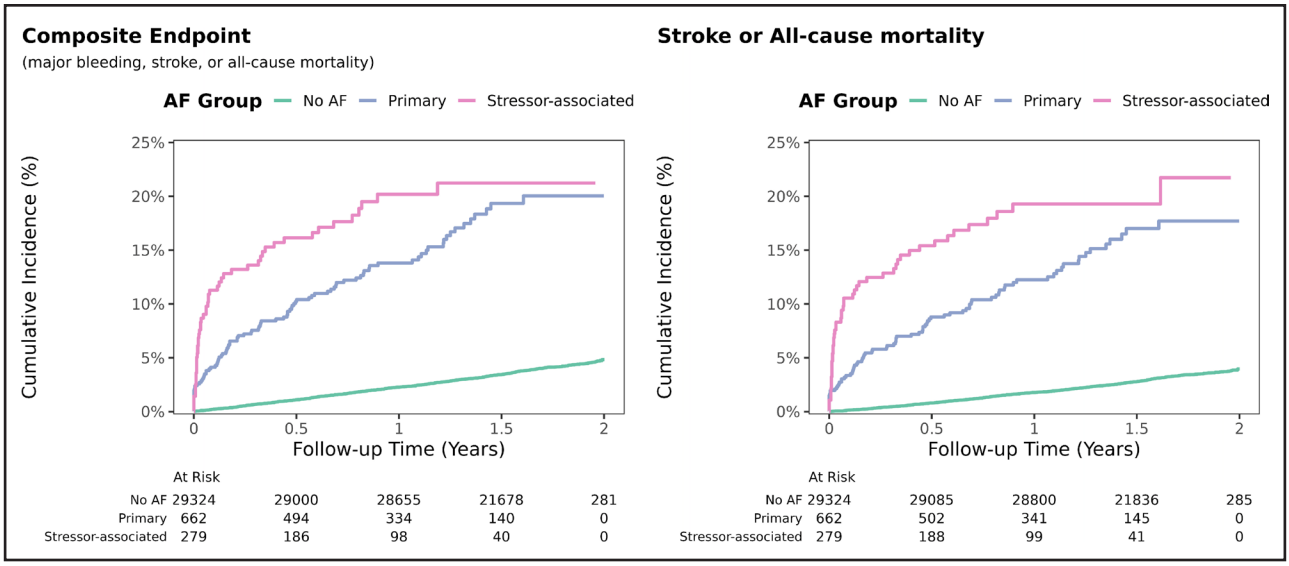


**Figure 3. Cumulative incidence of the composite endpoint and stroke or all-cause mortality by atrial fibrillation (AF) group.** Shown is the cumulative incidence of the composite end point (**left**) and stroke or all-cause mortality (**right**) stratified by AF group. A cumulative incidence curve depicts the time to first event for the composite end point (major bleeding, stroke, or all-cause mortality) and stroke or all-cause mortality among individuals with no AF, primary AF, and stressor-associated AF. At-risk counts are displayed at the **bottom**.

events, we compared risk factor profiles for primary versus stressor-associated AF, quantified OAC use after incident AF diagnosis, and evaluated associations between stressor-associated and primary AF with subsequent adverse outcomes. We found that stressor-associated AF is common among older adults, accounting for nearly one-third of new AF diagnoses. Risk factor profiles appeared to be largely similar between stressor-associated and primary AF. The incidence of adverse outcomes was substantial after an incident AF diagnosis of either type, occurring at a rate of approximately 20% at 2 years. Despite this, OAC use was lower among patients with stressor-associated AF, even among individuals at elevated stroke risk.

Our findings support and extend previous studies reporting substantial rates of stressor-associated AF in the community. In a longitudinal analysis from the Framingham Heart Study, 31% of individuals with incident AF were classified as having stressor-associated episodes, closely aligning with our finding that 29% of incident AF cases were stressor associated.[3] Consistent with previous reports, we observed that infection, cardiac surgery, and noncardiac surgery were among the most frequent stressors.[3,4,25] However, important limitations of most previous work include use of purely retrospective designs, reliance on billing codes to ascertain stressor-associated AF and related outcomes, and a narrow focus on individual AF stressors. Here, we leverage a large trial population of primary care patients with prospectively collected and manually adjudicated stressor-associated AF and outcomes data to systematically evaluate risk factors, patterns of OAC use, and outcomes associated with stressor-associated AF.

Our findings have 3 key implications. First, established clinical AF risk factors had very similar associations with both primary and stressor-associated AF, suggesting that opportunities to optimize modifiable AF risk factors should be similarly effective for preventing stressor-associated AF. Our analysis demonstrates that cardiovascular risk factors, including age, heart failure, and hypertension, were strongly associated with both incident AF types and extend previous evidence demonstrating risk factor associations in isolated stressors (eg, after cardiac surgery).[9,10,12,22,26–28] Given the growing public health burden of AF, current guidelines emphasize risk factor modification (eg, blood pressure control and smoking cessation) to reduce risk of primary AF.[1,2] Our findings provide important evidence showing that such approaches should extend to stressor-associated AF. Although risk factor profiles were generally similar, we did note some differences, including stronger associations between age and weight with primary AF versus male sex and diabetes for stressor-associated AF. Although future work is needed to better understand distinct pathophysiological mechanisms underlying stressor-associated AF,[29,30] our findings suggest that glucose homeostasis may play an important role[4,31,32] and that certain groups (eg, men) may be subject to higher risk.[4,31] Because the precipitants associated with stressor-associated AF are commonly foreseeable, opportunities may exist to optimize AF risk in anticipation of an upcoming stressor exposure (ie, in advance of noncardiac surgery), particularly among individuals with multiple AF risk factors. Notably, we were unable to assess the associations of certain AF risk factors, including socioeconomic status, dietary intake, and physical activity, which merit

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future investigation in the context of stressor-associated AF.[33–35] Additionally, despite guideline recommendations supporting prophylactic pharmacological interventions (eg, β-blockers, amiodarone) to prevent stressor-associated AF in specific settings (eg, after cardiac surgery), such therapies are underused.[1,2,36] Improved recognition of individuals at risk for stressor-associated AF may lead to more effective prevention.

Second, our findings reveal differences in OAC initiation according to AF group, highlighting potential opportunities to improve management of stressor-associated AF. Although evidence remains insufficient to uniformly recommend OACs after stressor-associated AF, contemporary guidelines advise consideration of OAC initiation, particularly for patients at elevated thromboembolic risk.[1,2,5] However, we found that, even at the highest levels of stroke risk, individuals with stressor-associated AF were ≈25% less likely to be treated with OACs compared with their primary AF counterparts. Although stressor-specific factors (eg, postoperative bleeding risk) may justifiably contribute to lower OAC use, rates were consistently lower across all stressors. Lower rates of OAC use in stressor-associated AF have similarly been reported in previous studies focusing on stressor-specific cohorts at medium to high stroke risk.[14–16] Lower OAC use is likely multifactorial, with potential contributions including therapeutic inertia stemming from historical perceptions that stressor-associated AF is self-limited and benign,[37] concerns about elevated bleeding risk in certain populations (eg, postoperative sepsis),[13,38–40] absence of validated tools to stratify risk of recurrent AF and thromboembolic outcomes,[7,16,40–42] and mixed evidence linking OACs to improved cardiovascular outcomes such as stroke and all-cause mortality.[39,43–49] Overall, our findings highlight a critical need to develop better risk stratification tools to guide informed application of OACs and other preventive interventions in stressor-associated AF.

Third, individuals who develop incident AF, whether primary or stressor associated, merit prioritization for preventive measures to lower the risk of adverse cardiovascular outcomes. Consistent with a seminal analysis of the Framingham Heart Study, we observed that the cumulative incidence of stroke, major bleeding, and all-cause mortality was markedly higher among individuals with incident AF of either form (≈20%) compared with those without (3%). Our findings therefore highlight the importance of recognizing incident AF as a risk signal for future adverse outcomes, meriting identification and management of modifiable risk factors (eg, hypertension, smoking cessation) as well as OAC use when appropriate. To that end, we did observe lower risk of adverse outcomes with OAC exposure, with no evidence of heterogeneity by AF type. Of note, although rates of adverse outcomes were similarly high with primary and stressor-associated AF, optimal management of stressor-associated AF is comparably poorly understood, with fundamental uncertainties with respect to optimal strategies for rate and rhythm control,[1,2,50] identification of candidates for OACs,[1,2,41,51,52] and appropriate surveillance for AF recurrence, which occurs in up to 50% of cases.[1–4,7,10] Whether emerging evidence supporting early rhythm control to improve cardiovascular outcomes in primary AF extends to stressor-associated AF events also remains unclear.[53] Future studies are warranted to improve risk stratification for adverse outcomes after stressor-associated AF and assess the extent to which specific measures to improve outcomes in AF extend to the stressor-associated AF context.

Our study has several limitations. First, although VITAL-AF enrolled 16 primary care practices across a major metropolitan area in the northeastern United States, the findings remain potentially subject to geographic and demographic specificity. Second, VITAL-AF was not specifically powered to detect differences in clinical end points among AF types, and analyses of some individual component outcomes had limited precision because of modest event counts. Despite this limitation, stressor-associated AF was clearly associated with a significantly higher risk of adverse outcomes compared with no AF, with rates comparable to primary AF. Third, OAC ascertainment relied on EHR medication lists, which may be subject to misclassification and preclude assessments of adherence or persistence. Fourth, outcomes were ascertained from the EHR, which would not have captured the events outside of our system. Because participants of the VITAL-AF trial receive primary care in our system as a function of eligibility criteria, we submit that such misclassification should be minimal. Fifth, although we submit that our ascertainment of stressor-associated AF using manually validated outcomes is a major strength of our study, we acknowledge that the absence of routine rhythm monitoring may have led to misclassification of low-burden paroxysmal AF and also precludes complete assessment of recurrent AF or AF burden. For similar reasons, we are unable to quantify the exact duration of stressor-associated AF episodes, which may impact management decisions and subsequent outcomes. Sixth, because stressor-associated AF requires incidence of both AF and a stressor, our risk factor analysis cannot distinguish the extent to which observed associations reflect risk of stressor-associated AF versus risk of the stressor itself (eg, acute myocardial infarction). However, we submit that collapsing across highly varied stressors reduces the likelihood that observed associations are driven by associations with specific stressors. Seventh, although appreciable rates of subsequent AF detection support the contribution of stressor-associated AF to the substantial rate of adverse outcomes observed, our study remains subject to residual confounding and cannot establish causality. Eighth, some cases of stressor-associated AF may represent initial diagnoses of primary AF facilitated by monitoring related to the occurrence

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of the stressor (eg, in-hospital telemetry).[5,52] This is an inherent characteristic of clinically defined stressor-associated AF, and reliable identification of such cases would require the presence of continuous monitoring before recognition of clinical AF. However, we submit that a substantially lower rate of subsequent AF detection in the stressor-associated AF group (but at rates consistent with previous reports) supports our classification of stressor-associated AF as a distinct entity.

In a trial population of >30 000 primary care patients ≥65 of age with adjudicated AF outcomes, we demonstrate that stressor-associated AF accounts for nearly one in 3 incident AF diagnoses and exhibits both similar risk profiles and associations with future adverse outcomes as primary AF. Despite comparably high rates of adverse outcomes, patients with stressor-associated AF are less likely to receive OAC, even in the context of elevated stroke risk. Our findings highlight the importance of efforts to prevent and manage stressor-associated AF, identify a critical unmet need to develop improved risk stratification tools to guide preventive interventions in stressor-associated AF, and motivate prospective trials to establish optimal care pathways to prevent and treat stressor-associated AF.

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### Supplemental Material

STROBE Checklist
Supplemental Methods
Tables S1–S7
Figures S1–S5

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